

HYDRA-CORE v3.1 · Industrial HMI & Telemetry

High-precision operator control interface designed for heavy industrial hydraulic machinery, telemetry monitoring, and safety operations.

Role: Designing Engineer | Timeline: Jun 2023 · Jan 2025 | Tools: Figma, Industrial Research, Telemetry Design

01 Executive Summary & Problem Statement

Heavy hydraulic machinery operators in manufacturing facilities encounter high cognitive load when operating multi-axis hydraulic presses and fluid power equipment. Harsh industrial lighting, factory vibrations, and dense analogue dials often delayed fault identification.

The objective of HYDRA-CORE v3.1 was to replace fragmented hardware switches and dials with an intuitive, touch-optimized Human-Machine Interface (HMI) displaying live pressure, flow rate, temperature, and emergency states with zero ambiguity.

02 User Research & Factory Operator Inquiry

Conducted on-site contextual inquiries with 14 machine operators and plant maintenance engineers across multiple shift rotations.

Key Finding 1: Operators need single-glance status recognition from up to 3 meters away under varying glare conditions.

Key Finding 2: Critical alarm states must override routine parameter adjusting with high-contrast visual cues and haptic acknowledgement.

Key Finding 3: Cycle parameter configuration needed to be condensed from 8 physical menus into a single touch workflow.

03 Design System & Ergonomic Visual Architecture

Built a comprehensive Figma component library containing 150+ reusable tokens, dial states, telemetry charts, and modal warnings.

Applied WCAG AAA contrast guidelines specifically tailored for industrial TFT displays and anti-glare touchscreens.

Designed touch target sizes adhering to DIN EN ISO 9241-110 standard (minimum 48x48px hit areas for gloved operator inputs).

04 Validation & Quantitative Business Impact

35% reduction in operator emergency reaction times during simulated pressure anomaly tests.

100% of operators successfully configured cycle routines without supervisor assistance on day one.

Reduced training onboarding time for new machine operators from 3 weeks to 4 business days.